

Telco Customer Churn - Dataset & Project Proposal

Dataset Description

Our team selected the Telco Customer Churn dataset from Kaggle. The original dataset contains 7,043 customer records and 21 columns. It includes customer characteristics, account and service information, tenure, monthly and total charges, contract information, payment method, and the target variable **Churn**, which indicates whether a customer left the telecom company.

Project Objective

The goal of this project is to prepare the dataset for a machine learning problem that predicts whether a telecom customer is likely to churn. A churn prediction model could help a telecom company identify customers who may be at risk of leaving so that retention efforts can be directed toward those customers.

What We Will Accomplish

The dataset will be split in Python into 70% training data and 30% testing data. Exploratory data analysis (EDA) will be performed on the training data only. The team will clean missing values and duplicates, correct data types, encode categorical variables, perform feature engineering, address class imbalance, and demonstrate scaling and normalization. Matplotlib visualizations and sample outputs will be used to show the data before and after preprocessing. The testing data will remain separate from the training-data EDA and balancing process.

Expected Result

The final result will be a reproducible Jupyter Notebook that documents the preprocessing workflow and produces a cleaned, numeric, machine-learning-ready training dataset. This prepared dataset can then be used to train a classification model whose target is customer churn.

Dataset Source

https://www.kaggle.com/datasets/kapturovalexander/customers-churned-in-telecom-services?select=customer_churn_telecom_services.csv